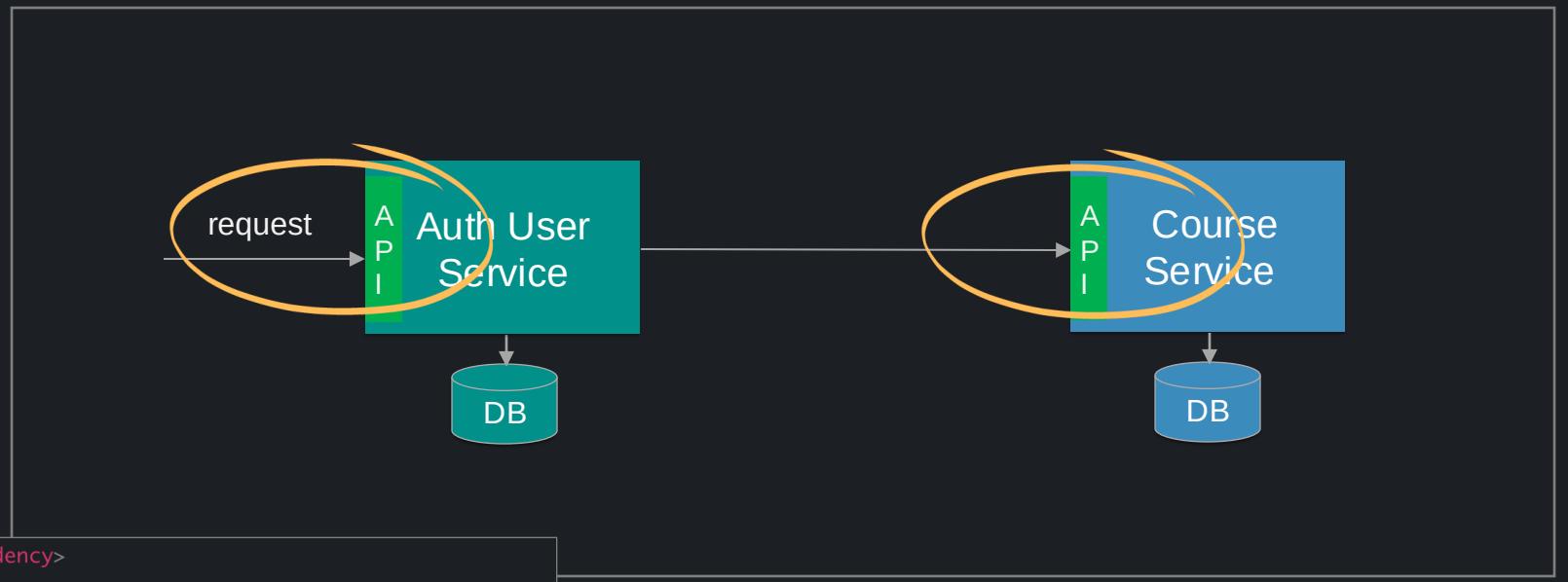
SPRING PROJECTS FOR MICROSERVICES ARCHITECTURE

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Spring Web MVC




```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
```

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Spring Validation

Annotations

- @Validated
- @NotNull
- @NotBlank
- @NotEmpty
- @Size
- @Email
- ...

```xml
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-validation</artifactId>
</dependency>
```

```java
@NotBlank
private String fullName;
@NotBlank
private String username;
@Email
private String email;
@Size(min = 6, max = 20)
private String password;
```

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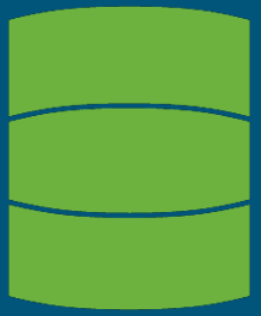


Spring HATEOAS

```xml
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-hateoas</artifactId>
</dependency>
```


```json
{
    "userId": "5930bac7-1555-4424-8702-c1778c974188",
    "username": "marcossilva",
    "email": "marcossilva@gmail.com",
    "fullName": "Marcos A Silva",
    "userStatus": "ACTIVE",
    "userType": "USER",
    "phoneNumber": "17988556512",
    "creationDate": "31-07-2024 17:18:28",
    "lastUpdateDate": "31-07-2024 17:18:28",
    "links": [
        {
            "rel": "self",
            "href": "http://localhost:8087/ead-authuser/users/5930bac7-1555-4424-8702-c1778c974188"
        }
    ]
}
```


hipermídias

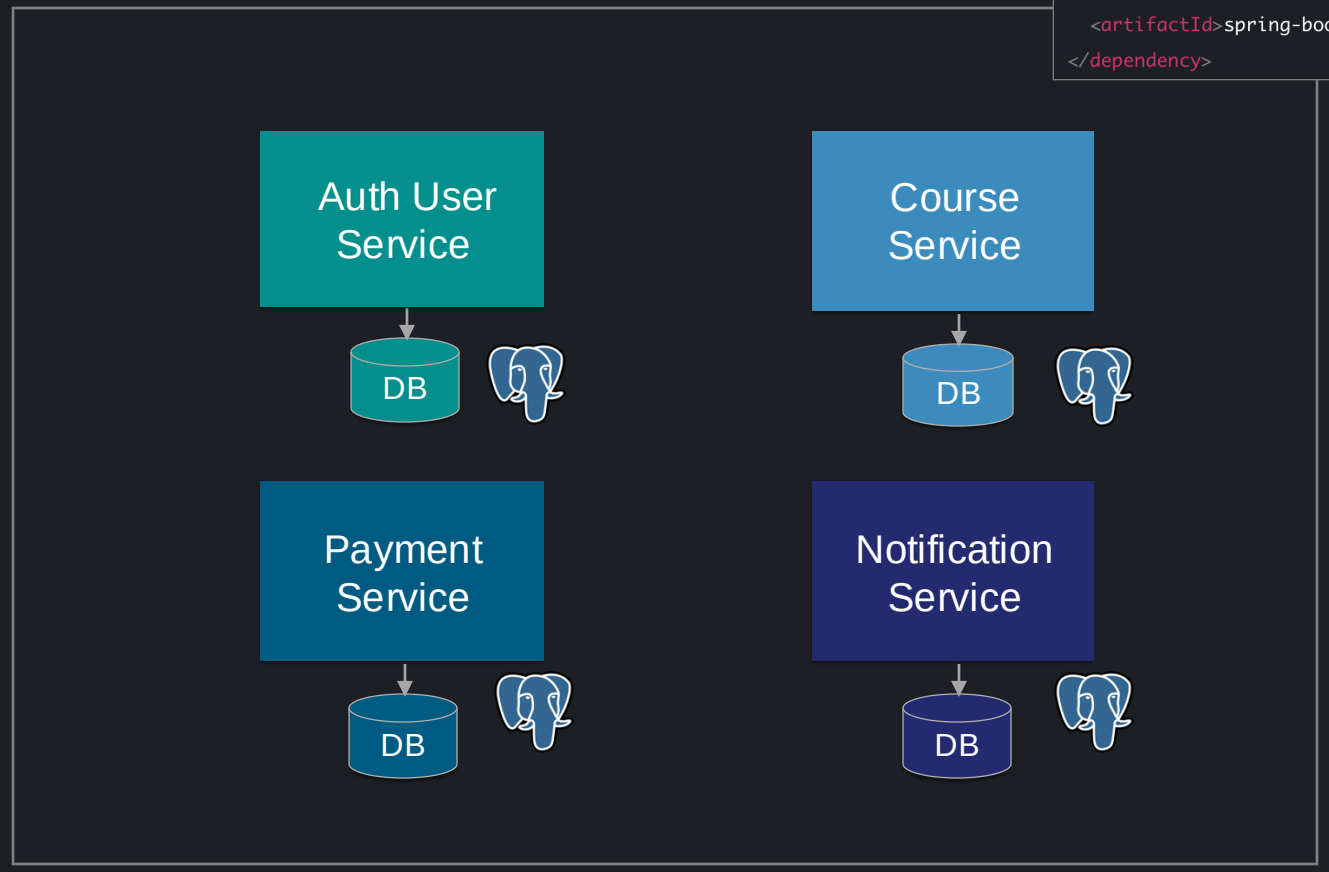


Spring Data JPA

JPA Methods

- findAll()
- findById(UUID id)
- save(Entity e)
- delete(Entity e)
- deleteById(UUID id)
- existsBy...()
- findAll(Pageable page)
- ...

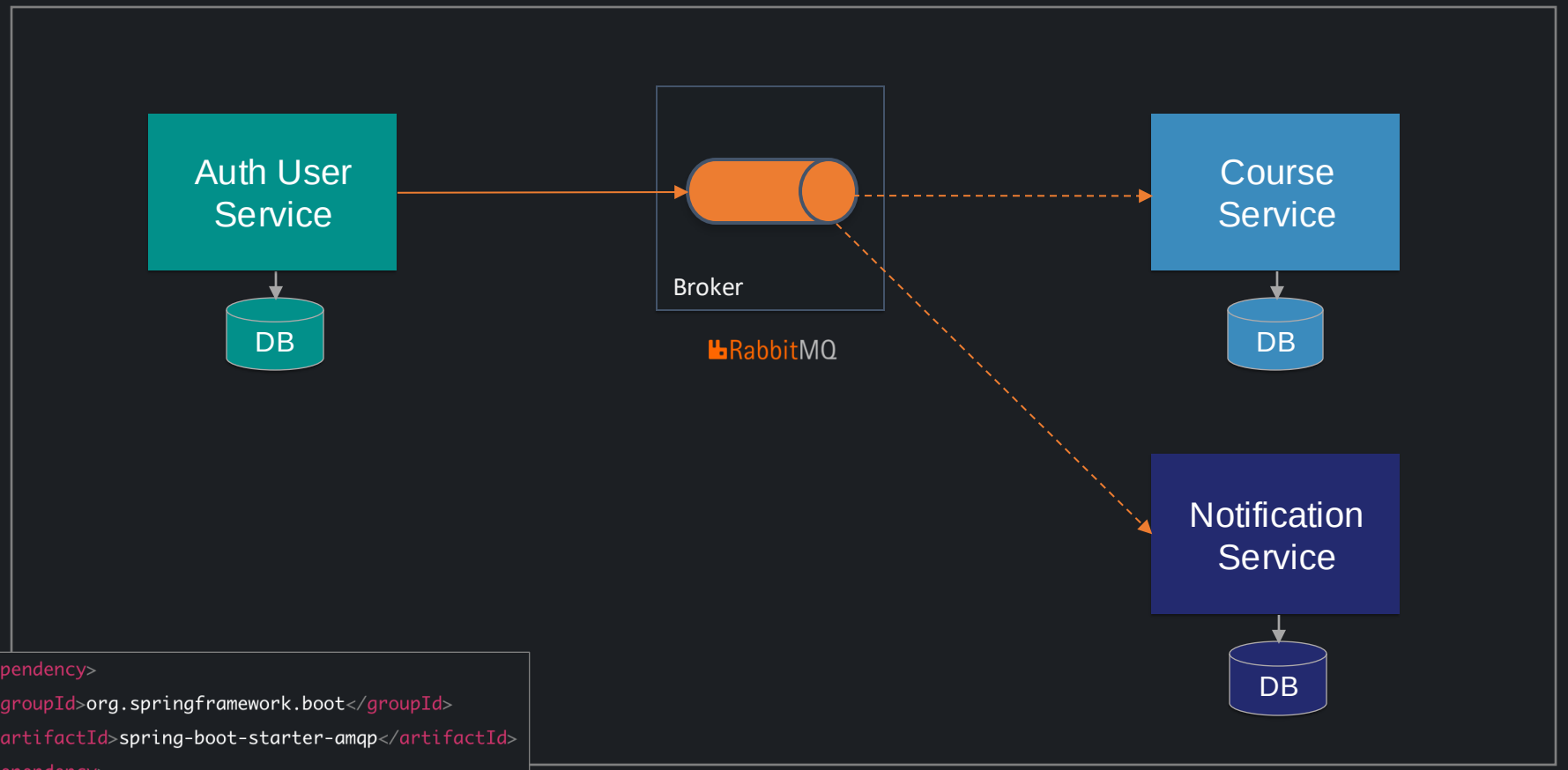
```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
```



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Spring AMQP



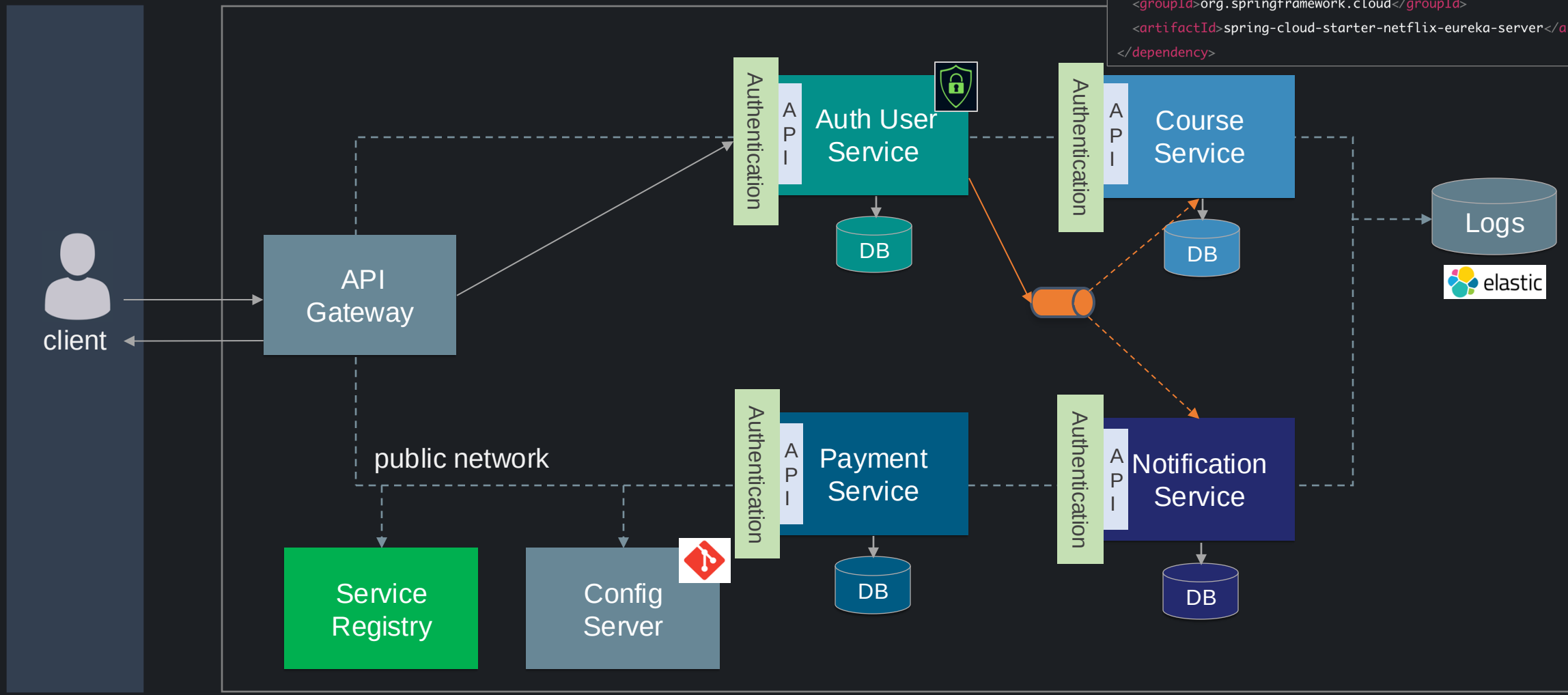

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-amqp</artifactId>
</dependency>
```

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Spring Cloud Discovery

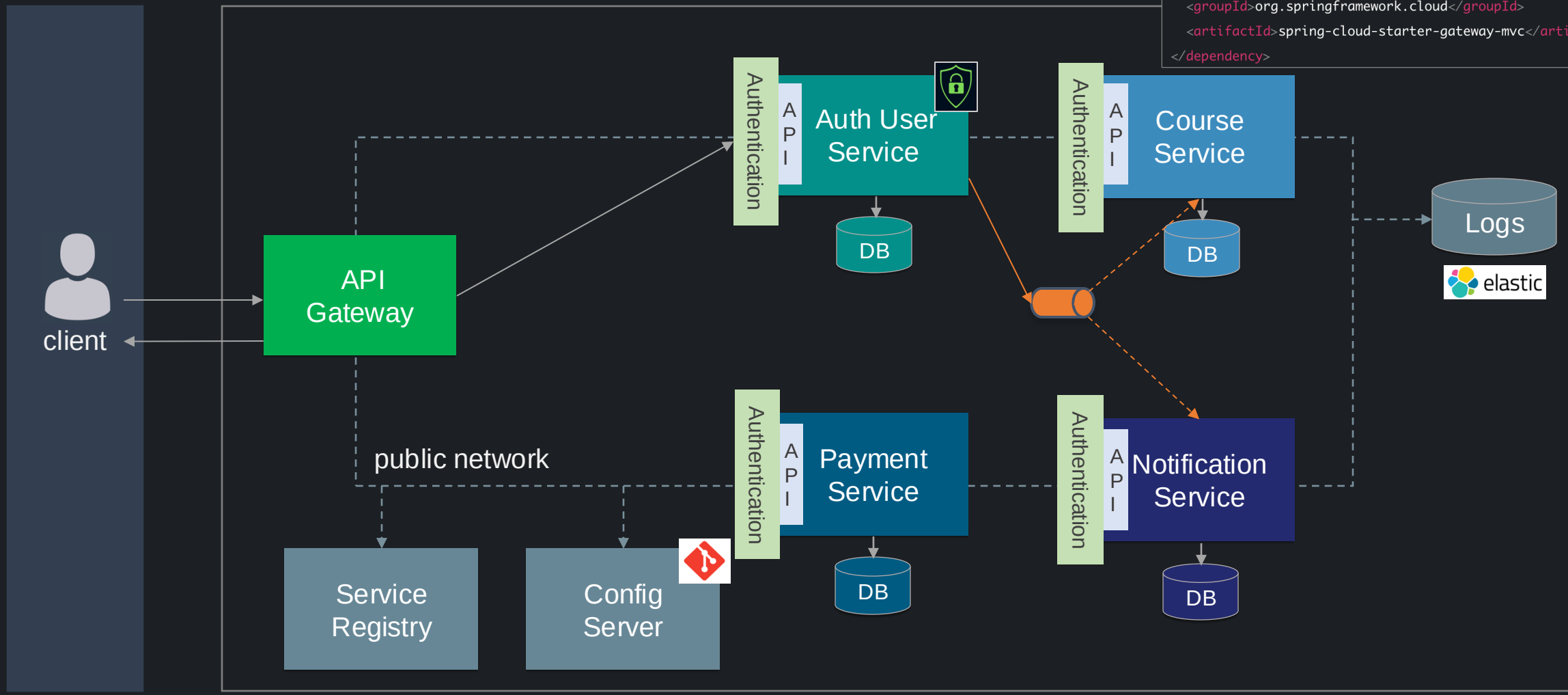
```
<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-starter-netflix-eureka-server</artifactId>
</dependency>
```



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Spring Cloud Gateway

```
<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-starter-gateway-mvc</artifactId>
</dependency>
```

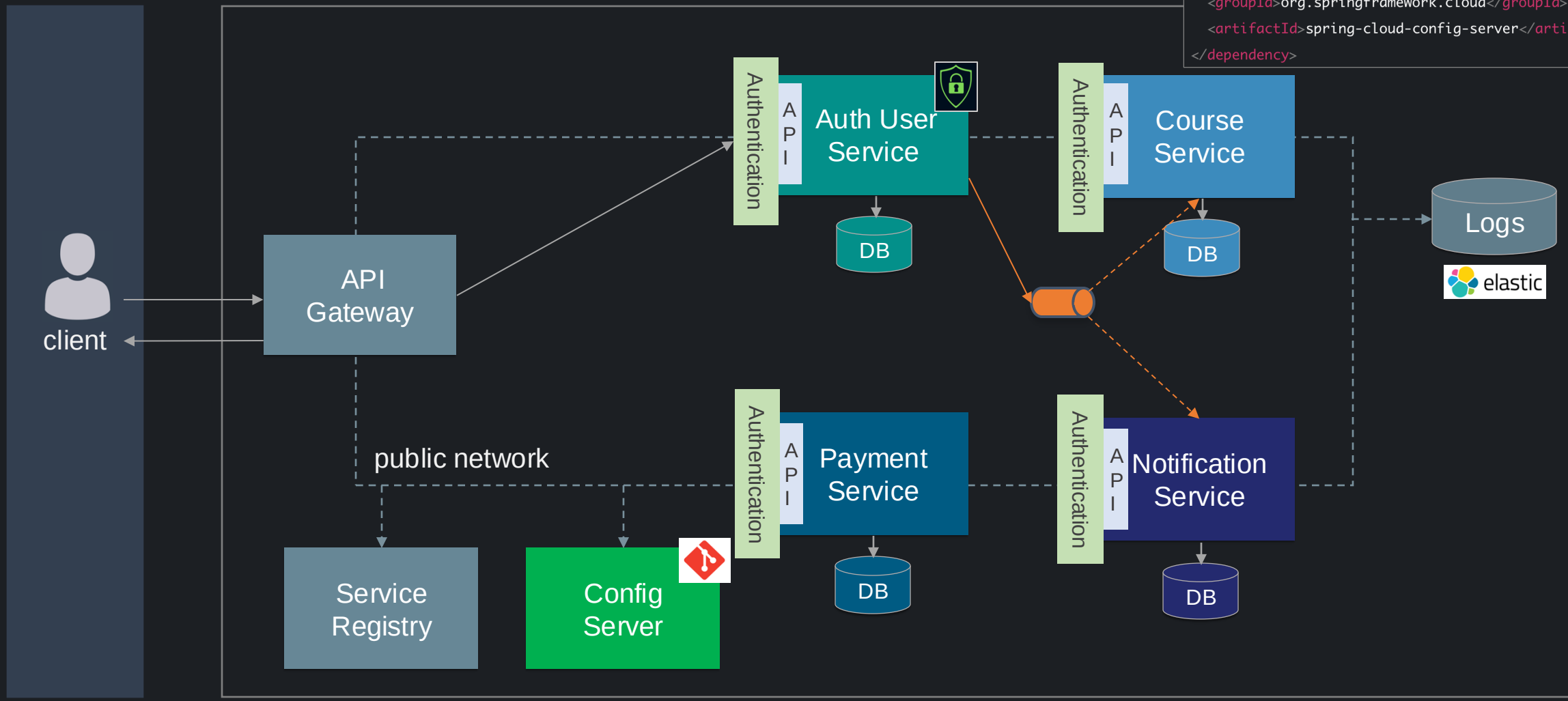


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Spring Cloud Config

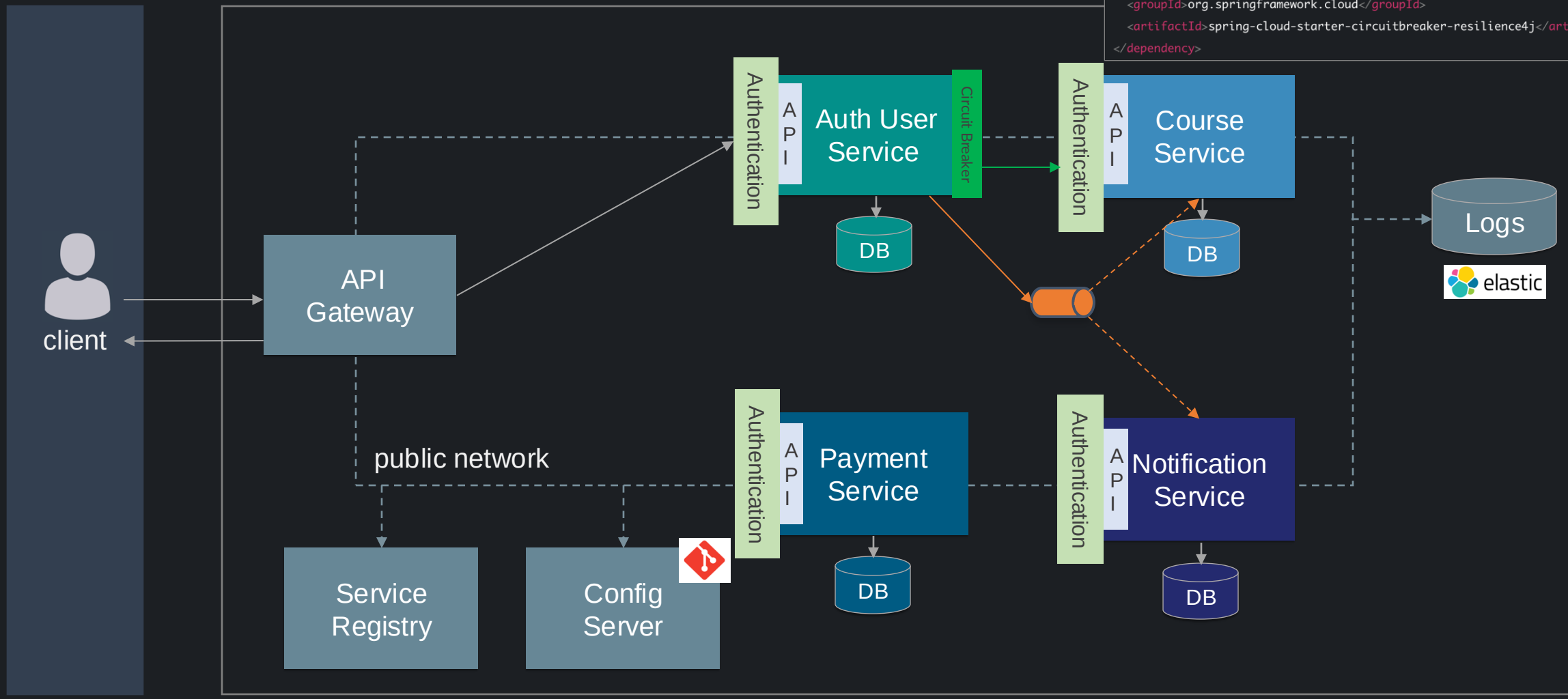
```
<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-config-server</artifactId>
</dependency>
```



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Spring Cloud Circuit Breaker

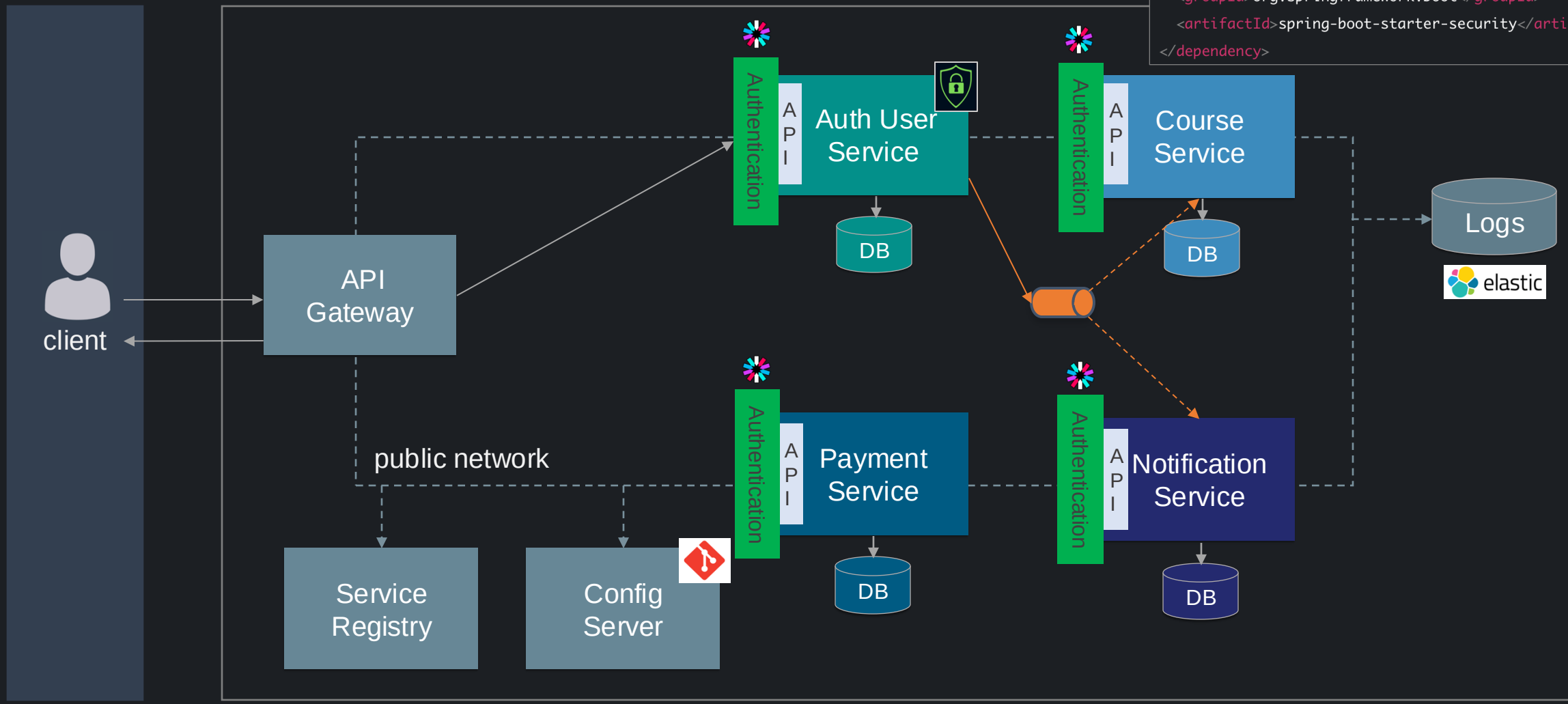
```
<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-starter-circuitbreaker-resilience4j</artifactId>
</dependency>
```



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Spring Security

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-security</artifactId>
</dependency>
```

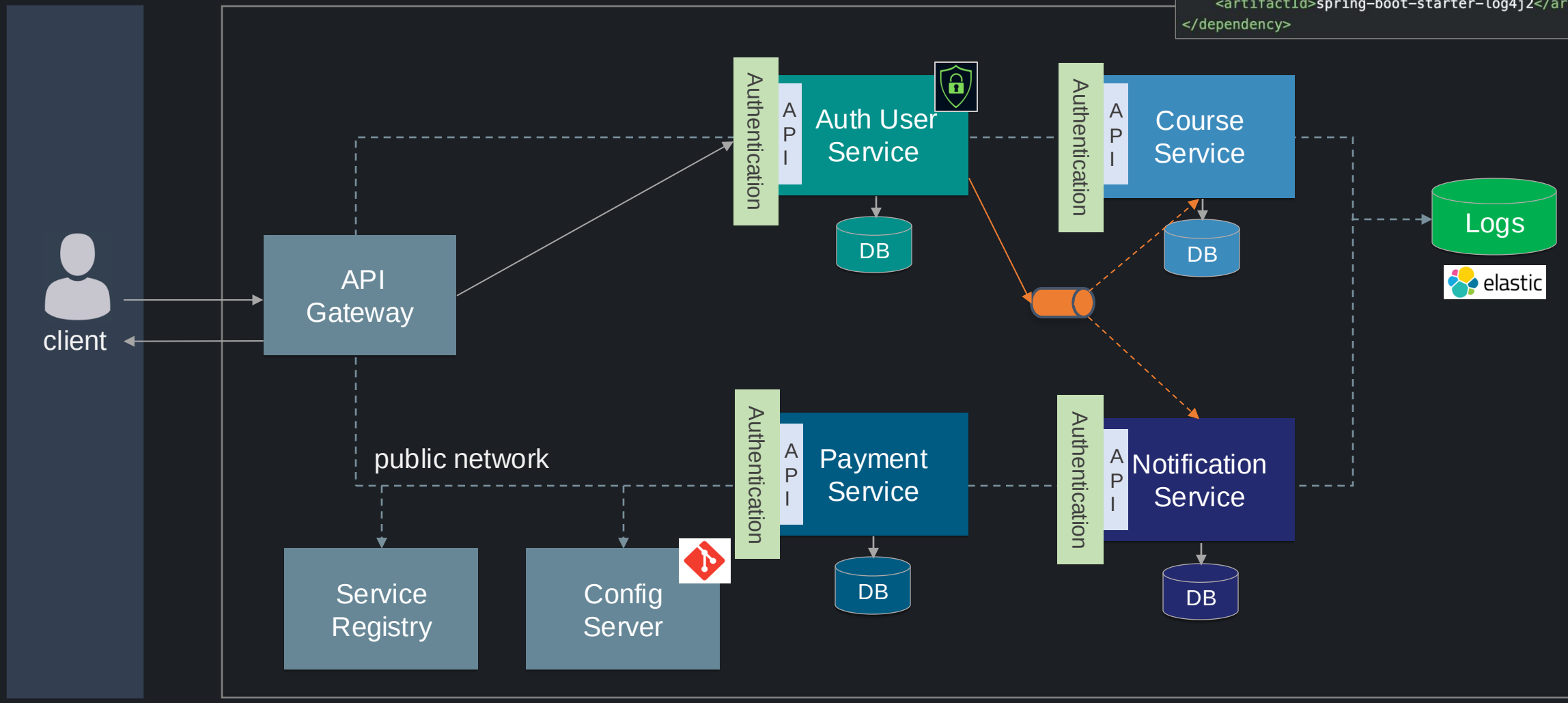


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Spring Logging

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-log4j2</artifactId>
</dependency>
```

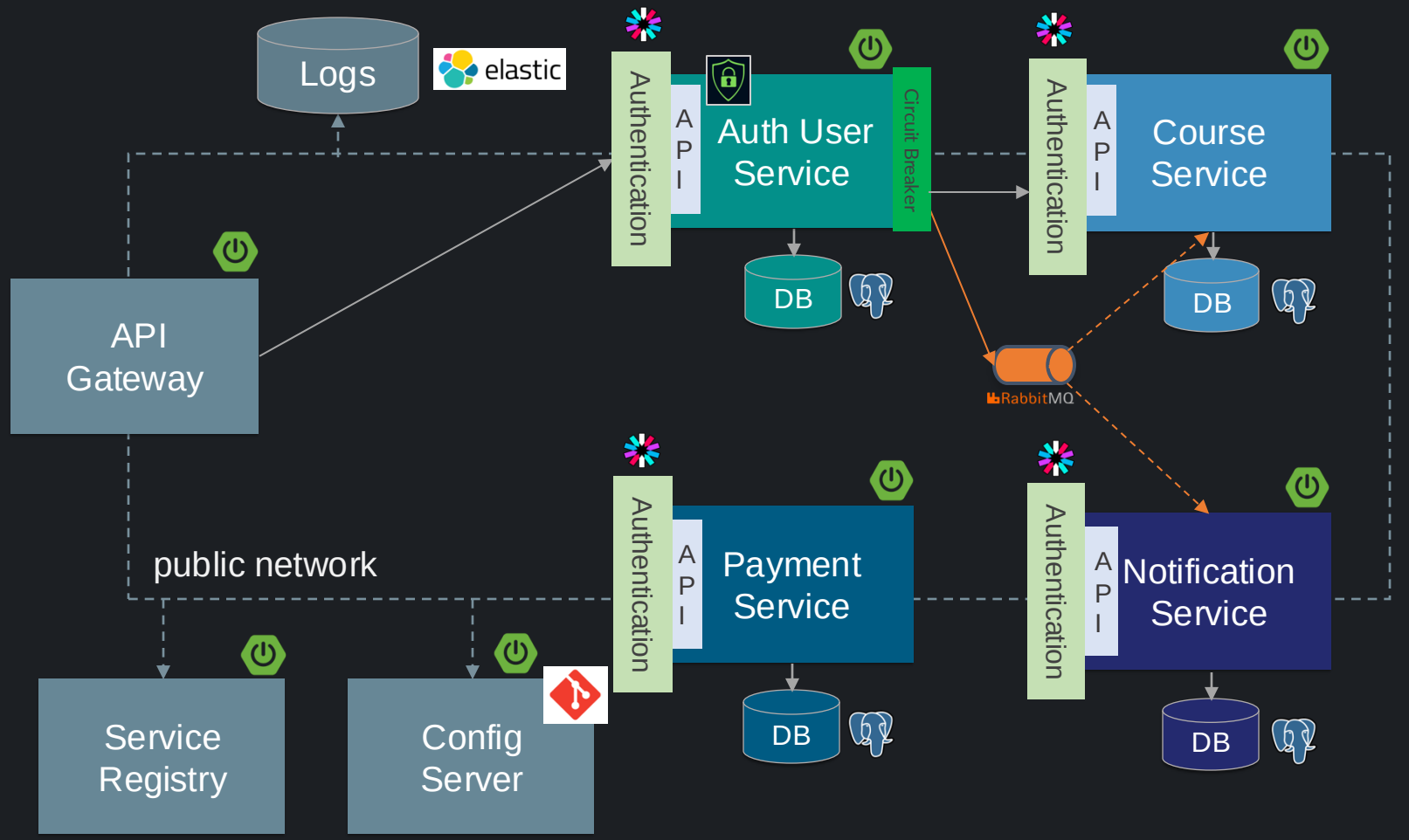


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Microservices Architecture with Spring - EAD

- Spring Actuator
- Spring Security
- Spring AMQP
- Spring Data JPA
- Spring Logging
- Spring Boot
- Spring Web
- Spring Hateoas
- Spring Validation



- Spring Cloud Circuit Breaker Resilience4j
- Spring Cloud Discovery Server
- Spring Cloud Discovery Client
- Spring Cloud Config Server
- Spring Cloud Config Client
- Spring Cloud Gateway

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